AI for Good Lab Hackathon: UAE – Mboa Labs – FlowPulse

Policy-grounded AI agents for urban mobility in Yaoundé, Cameroon (public services domain)

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Github link: https://github.com/Mboa-Labs/flowpulse (entry point) · knowledge base: https://github.com/Mboa-Labs/flowpulse-knowledge-base · live prototype: https://flowpulse2.duckdns.org

1. Introduction

Yaoundé, a city of more than 4 million people, plans its buses on outdated demographic data and reacts to congestion only once it has formed. **FlowPulse** is an agentic decision-support platform for the city's mobility managers. It turns aggregated mobile-network activity into an hourly **tension index** for each of the 7 communes, forecasts saturation 1 and 2 hours ahead, and lets AI agents propose concrete actions: reinforce a bus line, warn the police, inform passengers, or test a road closure before it happens.

Every proposal is checked by a **Policy agent** against a knowledge base of Cameroonian laws, national strategies, ITU standards and international references; the verdict cites the document and article. When no text applies, the agent does not invent a rule: it records a **regulatory gap**. No agent ever applies a decision: a human operator approves or rejects it, and every step is traced. The language model runs locally (Ollama), so no data leaves the server. The prototype is deployed and usable online.

2. Description of the use case

Problem. In Yaoundé, the Communauté Urbaine organises public transport, the bus operator runs the lines under contract and the police regulate traffic. None of them has a real-time view of where people are and where they are going. Today's practice relies on manual counts, fixed lines and reactive policing: decisions are late, undocumented and hard to justify.

Solution. FlowPulse chains five steps: *observe* (tension index per commune and hour), *forecast* (LightGBM models at 1 h and 2 h), *propose* (alerts with a contingency plan, bus reallocations computed from the reserve fleet), *verify* (Policy agent with cited sources) and *decide* (human approval, rejection with a mandatory reason). Five agents share the work:

Agent	Role	Uses the LLM
Watcher	Scans the communes; raises *dense* (≥ 75) or *saturated* (≥ 85) alerts, also when the forecast crosses the threshold	No (explainable rules)
Optimizer	Computes how many reserve buses would bring a line back to 70; submits the plan to the Policy agent	No
Policy	Retrieves the relevant passages, returns a verdict (compliant, compliant with reservations, non-compliant, not covered) with citations; logs gaps	Yes
Simulator	What-if sandbox: road, bridge or junction closure, event, reinforcement; traffic shift, travel time, measures verified by the Policy agent	Only for verification
Orchestrator	Conversational supervisor: chooses tools (city overview, zone traffic, fleet, regulations, reallocation proposal) and answers with sources	Yes

How it closes the gaps of existing practice. Alerts arrive before the jam, not after; each recommendation carries its legal basis; and the full chain (data, reasoning, sources, human decision) is reviewable. **Data.** No Cameroonian operator data is available yet. We processed the real **D4D-Senegal** dataset (Orange/Sonatel, Dakar, 2013) and transposed the hourly profiles of 7 Dakar districts onto the 7 communes of Yaoundé; volumes are not transposed. Roads, bridges, junctions, stadiums and markets come from OpenStreetMap; commune boundaries from geoBoundaries; the bus fleet is demonstration data. The interface states this transparently.

3. Use case requirement

- **Data (SRC, C).** Hourly antenna-level activity and origin–destination movements from at least one operator, delivered as aggregates only; a lawful basis and a written agreement between the operator and the Communauté Urbaine; bus lines, stops and fleet status from the operator.
- **Pre-processing (PP).** Aggregation by commune and hour before any computation; cells with fewer than 10 devices removed; no individual identifier stored or sent to a model; index computed against each zone's own 4-week high (P99), so that communes of different size remain comparable.
- **Model (M).** Forecast at 1 h and 2 h, evaluated on weeks never seen in training against simple baselines; alert threshold tuned to detect at least 75 % of saturations.
- **Policy (P).** Every action proposed by an agent is checked against the knowledge base before it is displayed; a verdict without a source is not allowed; an action is never applied without a named human decision.
- **Distribution and sink (D, SINK).** Role-based access (urban manager, bus operator, urban planner, administrator), JWT sessions, 30-minute inactivity logout, audit log exportable as CSV, interface usable on a phone in the field.
- **Infrastructure.** Runs on a single CPU-only server (Docker Compose, PostgreSQL, ChromaDB, Ollama with *llama3.2:3b* and *nomic-embed-text*); no external AI service.

4. Mapping with ITU-T Y.3172, the AI Readiness Report 2.0, and domain-specific policies

Y.3172 node	FlowPulse component	Readiness 2.0 dimension	Documents in the knowledge base
SRC	Telecom aggregates (D4D-Senegal), OpenStreetMap, geoBoundaries, bus network	Data and model marketplace; cross-domain correlation; digital infrastructure	Law 2010/013 on electronic communications (licence conditions, art. 9–10); Law 2024/017 on personal data
C	Ingestion and data-contract checks (pipeline/d4d.py)	Data: source and metadata	Law 2010/012 on cybersecurity; Malabo Convention (AU, 2014)
PP	Aggregation by commune and hour, threshold of 10 devices, tension index	Contextualisation; data harmonisation	Law 2024/017; gap GAP-002 (no anonymisation standard)
M	LightGBM forecasts at 1 h and 2 h, evaluated against baselines	Research; AI integration in the domain	ITU-T Y.3172; AI Readiness Framework 2.0

Y.3172 node	FlowPulse component	Readiness 2.0 dimension	Documents in the knowledge base
P	Policy agent (RAG, cited verdicts, gap register); human approval	AI and policy; collaboration with AI; human supervision	National Urban Mobility Policy (PNMU 2019); General Code of Decentralised Territorial Authorities (2019); SND30; Cameroon Digital Strategic Plan 2020
D	Orchestrator, alerts transmitted to operator, police or city hall	Deployment; strategy alignment between institutions	PNMU 2019 (roles of the Communauté Urbaine, ministries and police, p. 25–28)
SINK	Dashboards per role, audit log, mobile layout	Human interface; granular priorities	AU Continental AI Strategy 2024; Rwanda National AI Policy 2023
Sandbox	What-if Simulator: nothing applied, measures verified	Sandbox; AI for social inclusion	PNMU 2019 (planning tools, mobility observatory)

Knowledge base. 21 texts are inventoried in *sources.csv* with their official reference, authority, date and source URL; 10 are indexed (601 passages of national and international texts, 135 passages of ITU-T Y.3172 and the AI Readiness Framework 2.0, plus the gap register). Texts are split by article, then into passages of 256 words with a 30-word overlap, embedded with *nomic-embed-text* and stored in ChromaDB, following the ITU reference code (ITUAIReadiness). Each passage keeps its document, article, authority and date, so the agent can cite it. A 464-page telecom compendium was deliberately excluded: it would have made up 77 % of the passages and drowned the mobility texts.

5. Error scenarios

Scenarios A and B were run on the deployed prototype; scenario C describes the designed handling of an escalation. Events are released progressively, and information arrives incomplete and at different times.

Scenario A – Closure of Boulevard du 20 Mai, 07:00–10:00 (works or official ceremony).

1. *Planning.* The urban planner simulates the closure on the real OpenStreetMap trace. The loss of capacity is shared between the communes the boulevard crosses (Yaoundé III 47 %, I 36 %, V 17 %) and 60 % of its traffic shifts to neighbouring communes. Result: saturated communes go from 2 to 3, Yaoundé II gains 12 points, travel time +3 %, about 531,200 people concerned.
2. *Verification.* The Policy agent rates the closure *compliant* with the PNMU and proposes reserve buses for the saturated communes, each with its citations. Nothing is applied: it is a sandbox.
3. *Incomplete information.* On the day, the Watcher raises 5 alerts, each with a contingency plan. The operator applies two reinforcements but rejects the one for Yaoundé IV with the reason “two buses in maintenance at the depot”: the reason is stored in the audit log and the proposal is never applied silently.
4. *Controversy.* Police and city teams were not part of the decision. The agent can only recommend transmitting the alert, because no text defines who coordinates a real-time reinforcement (gap GAP-004).

Scenario B – Football match, 40,000 spectators at Ahmadou Ahidjo stadium, 16:00–20:00. The simulation adds 8 points to Yaoundé V (from 83 to 91, saturated). The Policy agent returns *non-compliant* with a justification about investment budgets, which does not

address the event. The verdict is shown with the passages actually retrieved; the urban planner sees the weak reasoning in the trace and keeps the decision. This is the intended behaviour: a small local model can be wrong, so its output is advice with evidence, never an automatic decision.

Scenario C – Escalation added in the spirit of the judges. A telecom operator offers finer, device-level location data to improve forecasts. FlowPulse refuses individual data by design: its pipeline only accepts aggregates. The legal question is routed to the Policy agent; as long as no indexed provision covers public-interest reuse of network data, its verdict is *not covered*, the case is written to the gap register and the human decision is referred to the data protection authority created by Law 2024/017 (gaps GAP-001, GAP-002 and GAP-007 below).

Technical evaluation (not a judging criterion, given for transparency). On November–December 2013 test weeks, the 1-hour forecast error is 3.1 points out of 100, versus 6.2 for "same hour last week" (−51 %); 71 % of saturations are announced and 70 % of alerts are correct. At 2 hours: 4.5 points (−27 %).

6. Policy gaps and standards gaps derived from the mapping exercise

Each gap below is recorded in the knowledge base (*registres/manques_reglementaires.csv*, collection *kb_manques*) with the texts examined. The agent adds new ones at runtime when no text covers a decision. These are domain-level gaps, each paired with a global policy example.

Category	Gap identified	Global policy example
Unclear responsibility	No text designates who may authorise an operator to share aggregated mobility data with a city, nor the procedure; licence conditions only require confidentiality (Law 2010/013, art. 9–10).	EU Data Act (Reg. 2023/2854), Chapter V, business-to-government data in exceptional need; GDPR recital 26 on anonymous data
Missing standard	No national technical standard to anonymise or aggregate mobile-network data (minimum devices per cell, re-identification test).	ISO/IEC 20889:2018; ITU-T X.1148 (de-identification framework for telecom providers)
Unclear responsibility	Continuous monitoring of urban mobility is "not specifically allocated by law" and "not performed" (PNMU, p. 25); the planned mobility observatory has no mandate to collect data.	EU Delegated Reg. 2017/1926 (national access points for transport data); France, Mobility Orientation Law 2019
Coordination	No mechanism brings together the Communauté Urbaine, the bus operator and the police for a real-time reinforcement; no traffic commission (PNMU, p. 28); operating contract not published.	EU Reg. 1370/2007 on public service obligations in passenger transport; UK Traffic Management Act 2004 (network management duty)
Missing standard	No open format is required for transit supply data (lines, stops, timetables, fleet).	GTFS; CEN NeTEx and SIRI, mandated by EU Delegated Reg. 2017/1926
Legal vacuum	No rule on AI decision support in public services: human oversight, record-keeping, liability. The National AI Strategy (July 2025) announces an AI law, but its full text is not published.	EU AI Act (Reg. 2024/1689), art. 12 and 14; Rwanda National AI Policy 2023; AU Continental AI Strategy 2024
Conflicting objectives	National strategies promote data-driven services, while Law 2024/017 imposes compliance without published guidance on aggregated data: public-interest pilots cannot tell whether to wait or proceed with safeguards.	Regulatory sandboxes: EU AI Act art. 57; UK ICO Regulatory Sandbox

7. Auditability, limitations and next steps

Auditability. Each answer, alert and simulation has a trace: agent, tool, inputs, outputs, duration and cited passages, shown with “See the reasoning”. Human actions (logins, acknowledged or transmitted alerts, approved or rejected reinforcements with their reason, accounts) go to an audit log exportable as CSV. Traces and decisions are persisted in PostgreSQL.

Limitations. The mobility profiles are transposed from Dakar 2013; the fleet is demonstration data; two key texts (Law 2024/017 and the Decentralisation Code) are scanned PDFs awaiting OCR; the 3-billion-parameter model sometimes writes weak justifications, which is why its verdicts stay advisory. **Next steps:** OCR and index the two texts, run a pilot with a Cameroonian operator under a data-sharing agreement, connect the bus operator's real fleet, and share the gap register with ITU study groups and national stakeholders.